

OSAT Investigation: Uncovering Yield Discrepancies

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Background: BRUIN is a world-class High Bandwidth Memory (HBM) supplier that has recently designed a new pair of memory products. These memory products are selling at an enormous premium currently, as a function of the global memory shortage. You have been selected by BRUIN management to investigate some suspicions about the yield of two of their most recently developed AI memory chips, Apache and Odyssey. The ASP of Apache is $175 and the ASP of Odyssey is $145 so yield loss of any magnitude is extremely costly.

- For assembly and test of Apache and Odyssey, BRUIN has signed a contract with a new OSAT called TestPAC in rural southeast Asia. TestPAC offers extremely competitive pricing as a function of their lower energy costs.
- TestPAC's energy is derived from a massive co-located solar farm.
- Cooling of the memory testers requires chilled water and TestPAC utilizes a new cooling system that they claim is also very energy efficient.
- TestPAC claims that use of their solar farm and low-cost water cooling system saves BRUIN $1.25 Million dollars in energy costs over a 32 day period.
- When evaluating TestPAC, BRUIN did not divulge the expected average yields of each product and contracted TestPAC to conduct 32 days of production testing across 24 testers on their test floor.
- 32 x 24 total lots of 10,000 units each were tested for each product, and the resulting yield matrices were sent to BRUIN for analysis.
- After analyzing the yields and touring the TestPAC facility, the lead Apache test engineer reported to BRUIN management that the yields should be extremely stable and even the mild yield variation being seen at TestPAC doesn't look quite right.

Suspicions: BRUIN Management has determined that the yields for Apache and Odyssey must be monitored on an ongoing basis by means of a straightforward AI dashboard. In addition to monitoring yields the dashboard must help resolve a few additional suspicions associated with the yields.

- Some of the test coverage for these devices, Apache in particular, requires extremely stable high-voltage, high-current stimulus. Even very small perturbations or dips in the tester power or dips in the chilled water temperatures may impact the test results.
- BRUIN suspects that the testers, which derive power from a TestPAC electrical system that is poorly conditioned and sourced by the solar array, are suffering from some power supply instability which is affecting the yields.
- Frequent weather changes and intermittent cloud cover cause the Solar Intensity to vary on a scale from 0-99. Zero represents lower temperatures and nearly complete cloud cover while 99 represents clear skies and high temperatures.
- Similarly, the humidity fluctuates significantly and when the humidity gets very high it has a big impact on the ability of the TestPAC chillers to keep water cool enough to pull sufficient heat away from the testers.
- Confidentially BRUIN has received information that some of their Apache chips are being sold on the black market which implies that at least one of the TestPAC test operators may be diverting Bin 1 devices which would affect the reported yield.

The Dashboard:

- Create a dashboard that BRUIN management can use to comprehensively observe the yields for multiple products. Add capabilities that you think make the dashboard most effective.
- The dashboard must be able to read multiple product files including a Solar Intensity and Humidity file that would help generate output highlighting the relationship between a chip's yield and the associated weather conditions.
- The dashboard must also be capable of flagging any operator(s) who can be associated with yield abnormalities.
- If yield abnormalities are detected, please indicate what the magnitude of the yield abnormality is on average and the cost of that suspicious deviation for the lots tested.

Problems to Solve:

- What are the respective average yields of Apache and Odyssey?
- Were any operator specific statistical abnormalities detected?
- If you did detect abnormalities, what financial costs would you assign to these operator abnormalities over the 32 days of testing?
- Did you find a relationship between the weather conditions and the yields of Apache and Odyssey?
- Would you say overall that the test partnership with TestPAC is financially preferable to an alternative test facility that uses costly but perfectly ideal electrical supply conditions for the testers and ideal climate-controlled humidity conditions for water chilling?
- What would your bottom-line recommendations be to BRUIN management about the TestPAC opportunity?